

AUTOMATIC CERTIFICATE GENERATION Using MATLAB

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MATLAB is a versatile tool that can automate many tasks. Presented here is a MATLAB code to generate certificates for workshops, conferences, symposiums, etc. Conventionally, educational institutions and companies use specialised tools for report cards and certificates generation on a large scale. This MATLAB code can be extended to generate analysis reports for large data sets as well.

Overview

The project overview is shown in Fig. 1. File name of the blank certificate (A) is used as input in the code. Details to be written on the certificate are obtained from an Excel sheet (B). The data is written over the blank certificate and many such certificates are generated by the MATLAB code (C). The certificates generated by the program can be saved in a folder with a unique file name (D).

Description

The three major steps involved in this project are shown in Fig. 2. Base image can be in any colour and should be in tif, jpg or png format. The image generated by the program retains properties of the base image. Source code given here only inserts text on the image. Data to be written can be in either .xls or .xlsx format. MS Excel is preferred as it's a powerful mathematical and statistical tool. It allows data to be internally computed and analysed within the file.

The position at which text is to be inserted can be identified using simple image editor tools such as MS Paint and Picture Manager. Pixels or picture elements are the most basic elements of an image. An image of 1024 × 768 resolution contains 786,432 pixels or nearly 0.79 megapixels. Pixel positions to be identified are as many as

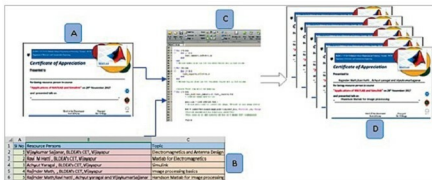


Fig. 1: Project overview

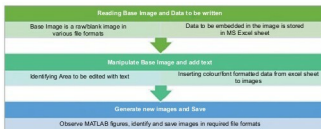


Fig. 2: Major steps and procedures in the project

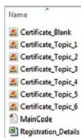


Fig. 3: MATLAB working environment

the number of texts and numbers to be inserted on the image. The project code can be very easily manipulated for similar image processing tasks.

The generated MATLAB figures or images can be saved into separate files or a single file as per requirement. By default, the code saves the generated image in tif format with a unique name. With simple code modifications, images can also be ported to a PDF file.

Working environment

Fig. 3 shows folder contents of the MATLAB working directory. MainCode.m, Certificate_Blank.tif and Registration_Details.xls files

should be saved in the same folder. MainCode.m is the MATLAB M-file containing the main code, Certificate_Blank.tif is the base image, and Certificate_Topic from 1 through 6 are certificates generated from the data in Registration_Details.xls. While running M-file, MATLAB should be permitted to change directory to the location where the main folder is saved.

Future scope

Analysis and presentation of large data sets is a tedious task in applications such as Big Data, IoT and sensors-actuators modelling. The project presented here can be extended and customised for analysis and reports generation in these applications. **EFY**



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EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com